

Technology Stack

Date	12 August 2026
Name	Chahat Chouhan
Team ID	Individual Project
Project Name	AI-Brand-Voice-Generator
Maximum Marks	2 Marks

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js with Tailwind CSS	Provides high component reusability, modular architecture, and rapid UI development. Declarative component structure simplifies real-time UI updates during AI content generation.
2.	Backend / Server-Side	Python (FastAPI / Flask)	Lightweight and high-performance framework optimal for asynchronous API requests, seamless AI/ML

			model integration, and handling prompt parsing logic.
3.	Database / Data Storage	SQLite / PostgreSQL	Delivers robust relational data storage for user account management, session history, and structured JSON schemas for saved brand voice guidelines.
4.	Cloud / Hosting / Deployment	Docker & Render / Vercel	Containerizes the application for consistent deployment across environments, providing automated horizontal scaling and high availability.
5.	Version Control & CI/CD	Git & GitHub Actions	Enables streamlined collaboration, automated unit testing, continuous integration, and seamless deployment workflows on main branch updates.
6.	Third-Party APIs / Other Tools	Google Gemini API	Powers core Natural Language Processing (NLP) capabilities, automated brand profile extraction, and context-aware marketing copy generation.

